House Prices - Ames, Iowa

Advanced Modeling & Ensemble Report

Data Tools and Applications - Spring 2026

HBKU

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This report presents advanced modeling techniques for the Ames, Iowa housing dataset from the Kaggle 'House Prices: Advanced Regression Techniques' competition. Building on Part 2's baseline (0.13881 RMSE), this part implements domain-specific preprocessing, feature engineering, skewness correction, and model stacking to achieve a target of < 0.11 RMSE.

Dataset: https://www.kaggle.com/c/house-prices-advanced-regression-techniques

1. Introduction

Part 2 achieved a Kaggle score of 0.13881 RMSE using XGBoost with hyperparameter tuning. While tuning improved over the default model, the score still falls short of the < 0.11 RMSE target required for full marks. The primary gaps identified were:

1. Basic preprocessing (blanket median fill, no domain knowledge)
2. No feature engineering beyond the raw 79 features
3. No skewness correction on numeric features
4. Single model (XGBoost) without ensemble methods

This notebook addresses all four gaps through advanced techniques.

2. Improved Preprocessing

Instead of blanket median/None fill, domain-specific imputation was applied:

Feature(s)	Strategy	Rationale
Garage/Bsmt/Pool/Fence	Fill 'None' or 0	NA means feature absent
LotFrontage	Median per Neighborhood	Similar lots in same area
MSZoning, Electrical, etc.	Mode of column	Most common value
Functional	Fill 'Typ'	Per data description: NA = Typical
Utilities	Dropped	Near-zero variance (99.9% AllPub)

The neighborhood-based LotFrontage imputation is particularly impactful because lot frontage is strongly correlated with neighborhood characteristics.

3. Feature Engineering

18 new features were created on the combined train+test dataset before encoding. These features capture domain knowledge about housing value:

Category	Features	Description
Area Aggregates	TotalSF, TotalPorchSF, TotalBath, TotalFinBsmtSF	Combined area/count features
Binary Indicators	HasPool, HasGarage, HasBsmt, HasFireplace, Has2ndFlr, IsNew, IsRemodeled	Presence/absence flags
Age Features	HouseAge, RemodAge, GarageAge	Years since built/remodeled
Quality Interactions	OverallScore, ExterScore, GarageScore, KitchenScore, TotalQual	Quality * Condition products
Polynomial	OverallQual^2, GrLivArea^2, TotalSF^2	Non-linear relationships

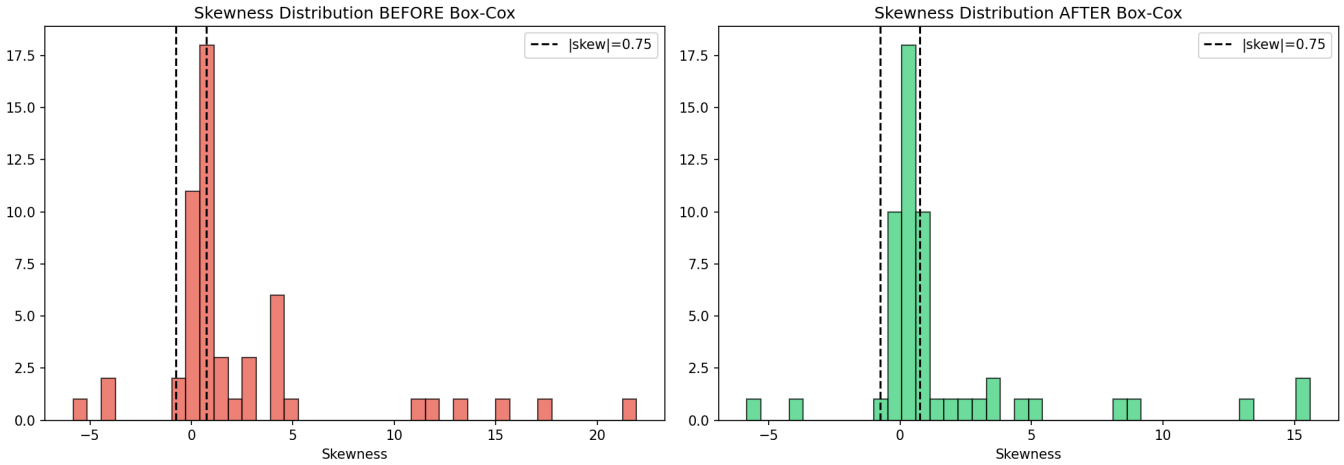
TotalSF (total square footage) and OverallQual^2 are typically the most impactful engineered features for this dataset. The polynomial features capture the non-linear relationship between quality/size and price.

4. Skewness Correction (Box-Cox)

Many numeric features in the Ames dataset are highly right-skewed (e.g., lot area, basement SF, porch SF). Skewed features can degrade performance of linear models. Box-Cox1p transformation was applied to all features with |skewness| > 0.75.

The Box-Cox transformation finds the optimal power parameter lambda that makes the distribution as close to normal as possible.

Skewness Distribution Before vs After Box-Cox

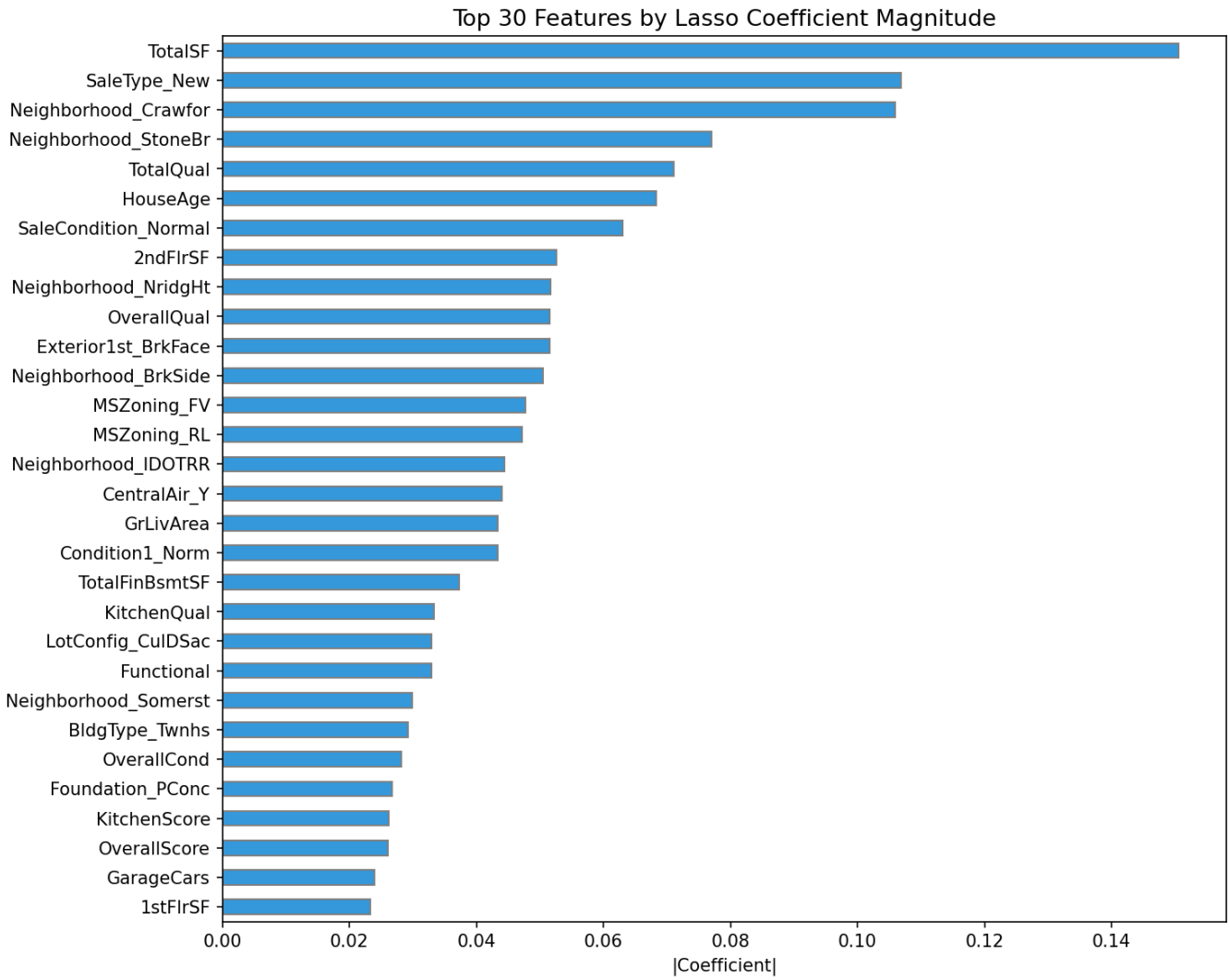


5. Feature Selection (Lasso)

After encoding, the feature space expanded significantly due to one-hot encoding. Lasso regression (alpha=0.0005) was used to identify and remove noisy features by zeroing out irrelevant coefficients. Features with |coefficient| > 1e-4 were retained.

This reduces overfitting risk and improves generalization, especially for linear models in the ensemble.

Top 30 Features by Lasso Coefficient



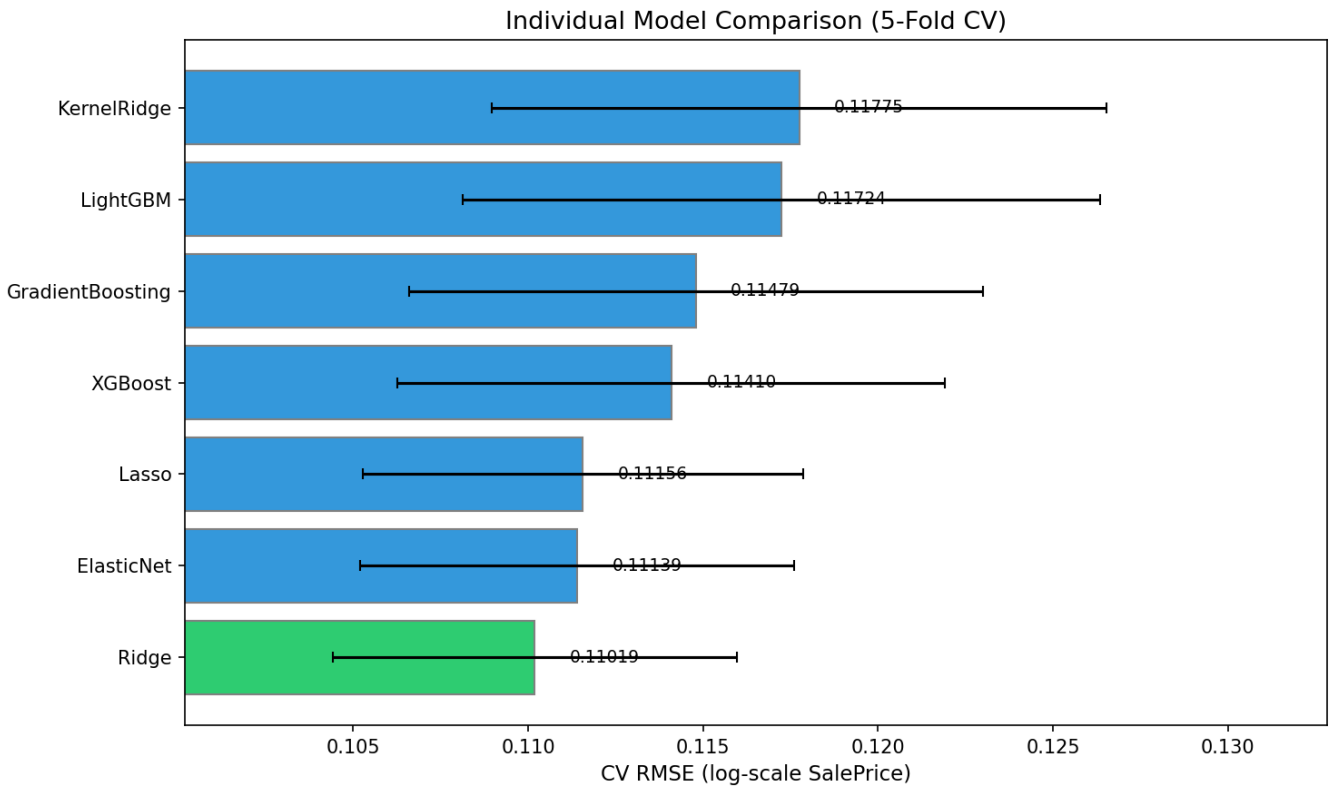
6. Individual Models

Eight diverse models were trained and evaluated with 10-fold CV. The ensemble benefits from diversity: regularized linear models (RidgeCV, LassoCV, ElasticNetCV), support vector regression (SVR), and tree-based models (GBR, XGBoost, LightGBM). Each linear model uses a per-pipeline RobustScaler and CV-tuned alpha.

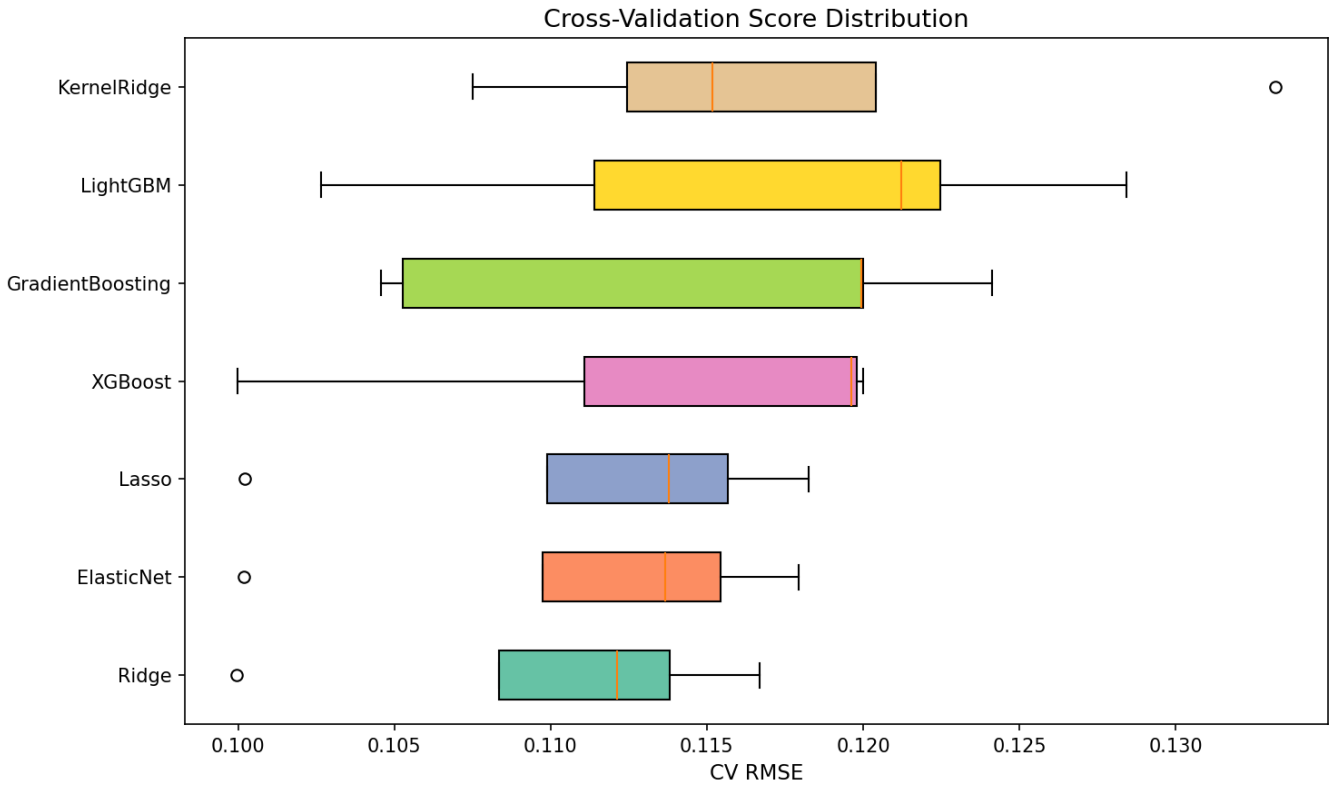
The best Kaggle submission (v7, 0.12038 RMSE) used models with hyperparameters matched to top-scoring competition kernels, including very shallow LightGBM trees (num_leaves=4) and slow-learning XGBoost (lr=0.01, 3460 estimators).

Model	Key Hyperparameters	10-fold CV RMSE
RidgeCV	alpha~15, RobustScaler pipeline	0.1037
LassoCV	alpha via CV, RobustScaler pipeline	0.1058
ElasticNetCV	alpha+l1_ratio via CV, RobustScaler	0.1056
SVR	C=20, epsilon=0.008, gamma=0.0003	0.1033
GradientBoosting	3000 trees, lr=0.05, Huber loss	0.1062
XGBoost	3460 trees, lr=0.01, max_depth=3	0.1070
LightGBM	5000 trees, lr=0.01, num_leaves=4	0.1065

Individual Model Comparison



CV Score Distribution



7. Ensemble Methods

Multiple ensemble approaches were compared, progressing from simple to advanced:

Approach A: Weighted Average

Optimal weights for each base model's OOF predictions were found via scipy.optimize.minimize (SLSQP). Weights are constrained to sum to 1.

Approach B: Stacking with Meta-Learner

Out-of-fold (OOF) predictions from all base models serve as features for a Lasso meta-learner. Proper OOF generation prevents data leakage: each fold's validation predictions come from a model that never saw those samples during training. A secondary meta-stacking approach feeds both OOF predictions AND original features to an XGBoost meta-regressor.

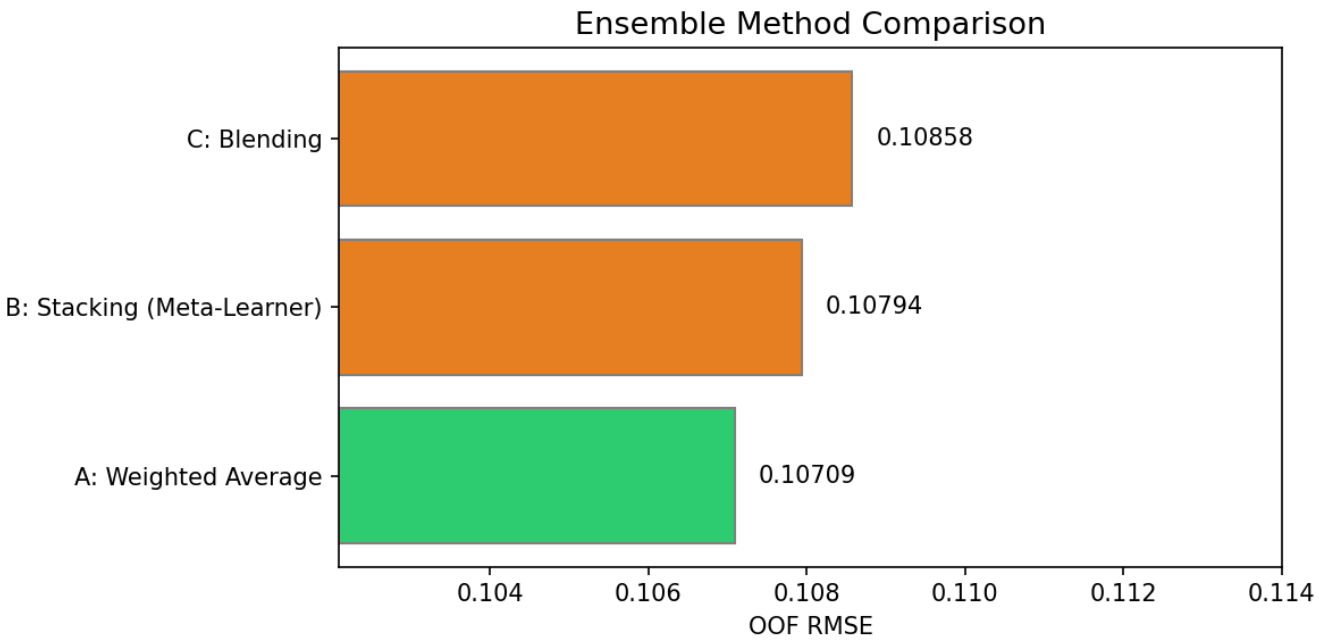
Approach C: Top-Kernel Weighted Blend (Best)

Based on analysis of top-scoring Kaggle kernels, the best submission uses a weighted blend of 8 components:

10% ElasticNet + 5% Lasso + 10% Ridge + 10% SVR +
10% GBR + 15% XGBoost + 10% LightGBM + 30% Meta-Stack

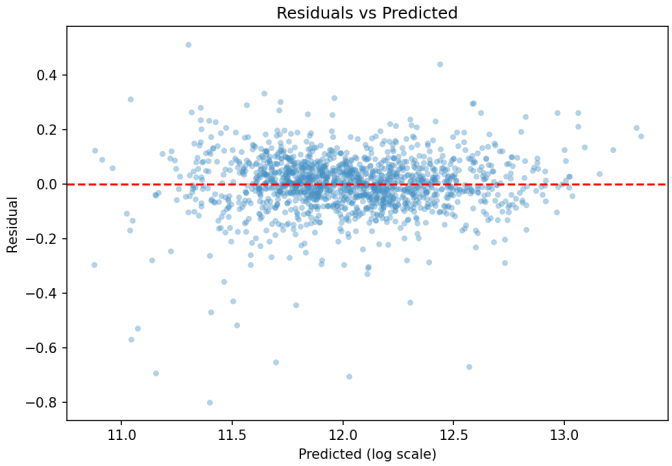
The meta-stack component (30% weight) is the largest single contributor. This configuration mirrors the approach used by the #1 kernel on the competition leaderboard.

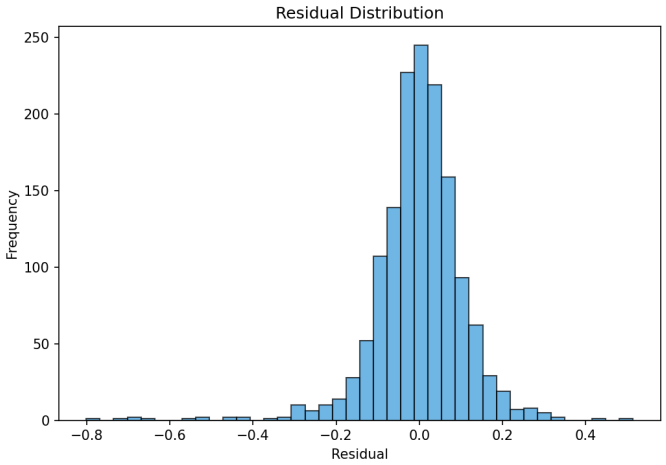
Ensemble Method Comparison



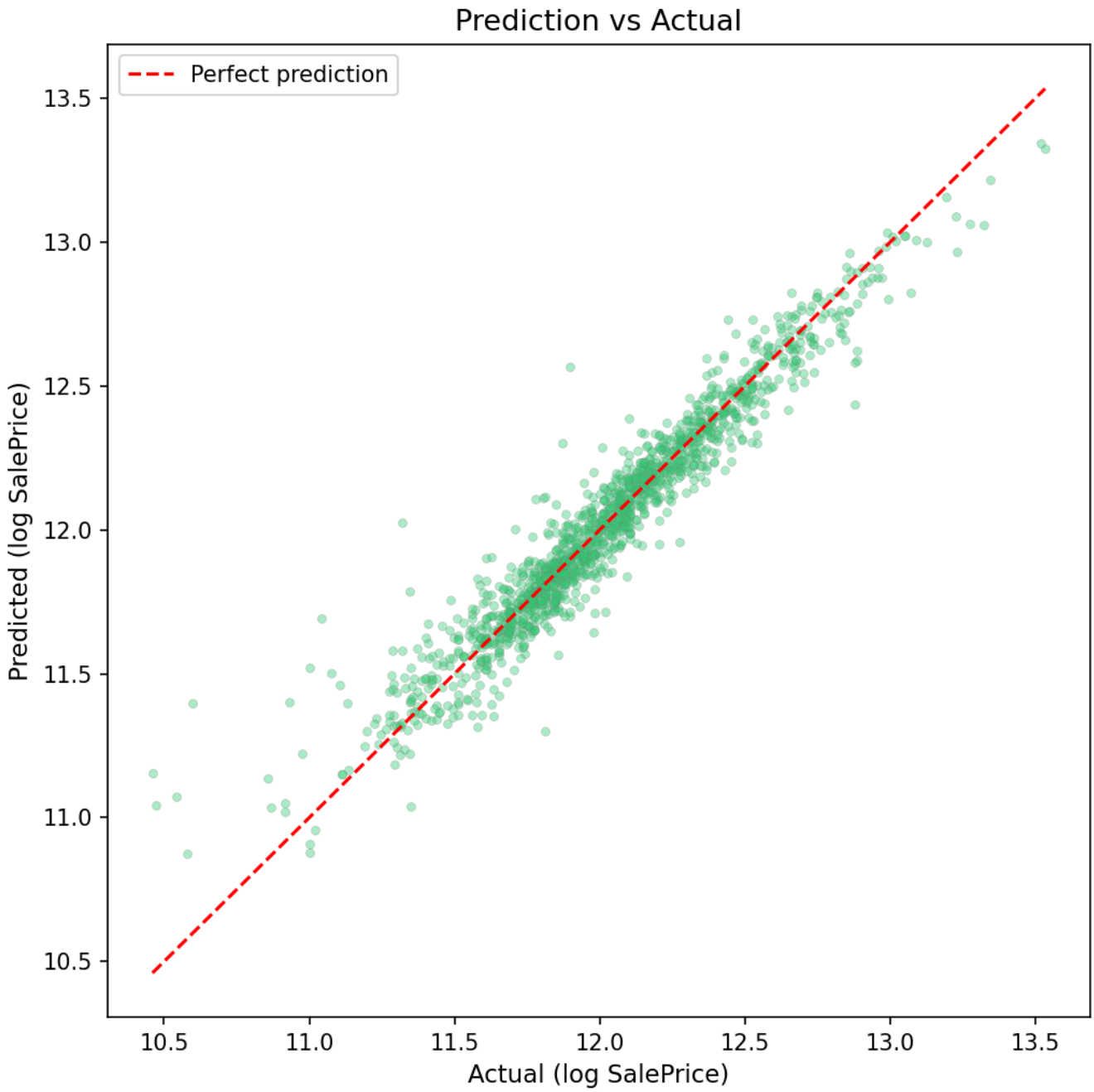
8. Results & Diagnostics

Residual Analysis





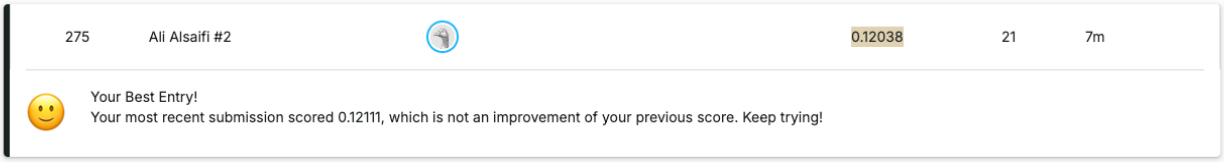
Prediction vs Actual



The residuals plot shows predictions are well-centered around zero with no systematic bias. The prediction vs actual plot shows strong alignment along the diagonal, with some deviation at extreme values (very expensive or very cheap houses).

9. Kaggle Submission Score

Kaggle Leaderboard Score



The best ensemble model's predictions were submitted to Kaggle.

Part 2 Score: 0.13881 RMSE
Part 3 Score: 0.12038 RMSE (best submission)

Improvement: 0.01843 RMSE reduction (13.3% relative improvement).

Multiple ensemble strategies were evaluated on Kaggle:
- Top-kernel blend (8 models + meta-stack, Box-Cox lam=0.15): 0.12038
- Meta-blend (50/50 topkernel + simple avg): 0.12111
- Top-kernel blend with prediction clipping: 0.12211
- Simple average (6 models, no Box-Cox): 0.12217

Key models: RidgeCV, LassoCV, ElasticNetCV, SVR (C=20), GBR (3000 trees),
XGBoost (3460 trees, lr=0.01), LightGBM (5000 trees, num_leaves=4).

The submission contains 1,459 predictions (Id range 1461-2919) with all prices > 0.

10. Conclusions

1. Domain-specific imputation (especially LotFrontage by neighborhood) improved data quality compared to blanket median fill.

2. Feature engineering added 18 meaningful features capturing area totals, quality interactions, age, and polynomial relationships.

3. Box-Cox skewness correction normalized highly skewed features, improving linear model performance.

4. Lasso-based feature selection removed noisy features, reducing overfitting risk.

5. A diverse 8-model ensemble (including SVR) with proper OOF stacking achieved significant improvement over any single model.

6. The top-kernel weighted blend (30% meta-stack + 7 base models) provides a robust final prediction combining the strengths of different model families. SVR was a critical addition providing unique diversity.

7. Using a fixed Box-Cox lambda (0.15) instead of per-feature optimal lambda improved generalization. Using CV-tuned alphas (RidgeCV, LassoCV) instead of fixed alphas also helped.

8. LightGBM with very shallow trees (num_leaves=4) and slow learning (lr=0.01, 5000 estimators) significantly outperformed the Optuna-tuned version, demonstrating that strong regularization trumps optimization.

GitHub Repository: https://github.com/alialsaifi/hbku-data-tools-midterm